# Hardware Review Report



Version 1.0
Espressif Systems
Copyright © 2023

# Table of Contents


1. Schematics....1

1.1. Power and Ground....1

1.2. RF....1

1.3. Clock....1

1.4. Reset/Enable....1

1.5. Sleep/Wake up....2

1.6. External RC....2

1.7. I/O....2

1.8. Flash/PSRAM....3

1.9. UART....3

1.10. Package....3

1.11. Others....4

2. PCB....5

1.12. PCB Check....5

1.13. Power and Ground....5

1.14. RF....7

1.15. Clock....8

1.16. Reset/Enable....8

1.17. External RC....8

1.18. I/O....8

1.19. UART Layout....8

1.20. Others....8


CONFIDENTIAL

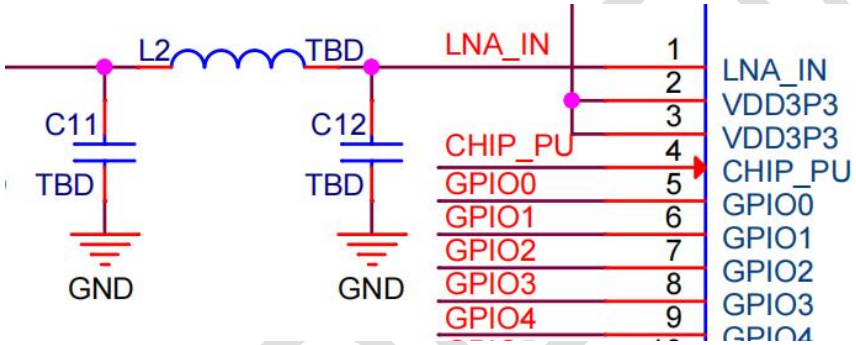
# 1. Schematics

## 1.1. Power and Ground

a. Please make sure the power supply current for ESP32-S3 is not less than 500 mA.

## 1.2. RF

a. Since S3 RF pin impedance point is not 50 ohms, please add a CLC matching circuit to tune S3 chip.




b. Adding a CLC matching circuit near the antenna is recommended to adjust the antenna. If the antenna impedance point can be guaranteed to be 50 ohms by simulation, then there is no need to add a matching circuit near the antenna.

c. Please be noted that all matching parameter values should be obtained after RF tuning process.

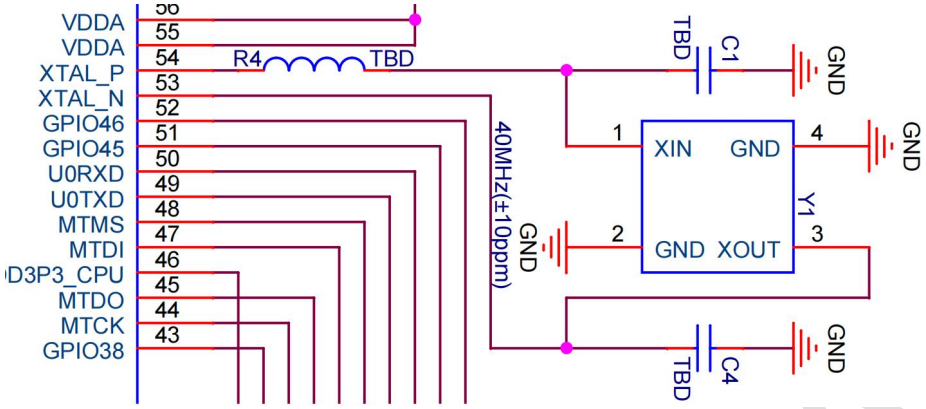
d. Why the layout is different from schematic in RF part?

e. If product will be affected by ESD during application or production process, it is suggested to reserve ESD protection close to antenna side.

## 1.3. Clock

a. Please add a series component at XTAL_P line, then use a 24 nH inductor as initial value for this series component.

CONFIDENTIAL



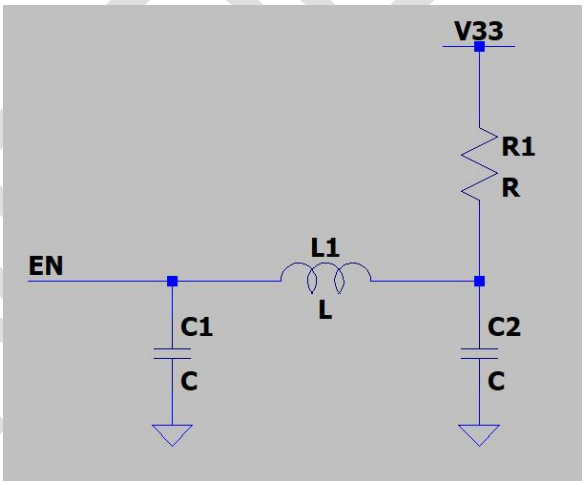

b. Please be noted that the final values of series inductor and parallel caps should be obtained after RF tuning process.

c. Please be noted the precision of 40 MHz should be at least 10 ppm.

## 1.4. Reset/Enable

a. Please be noted that RC at CHIP_PU pin should be adjusted with real power to meet the requirements of power-up and power-up sequence.

b. If space is allowed, it is suggested to reserve another CL close to CHIP_PU pin to improve EMC performance. Initially, do not stuff for C1 and use 0 ohm resistor for L1




## 1.5. Sleep/Wake up

a. NA

## 1.6. External RC

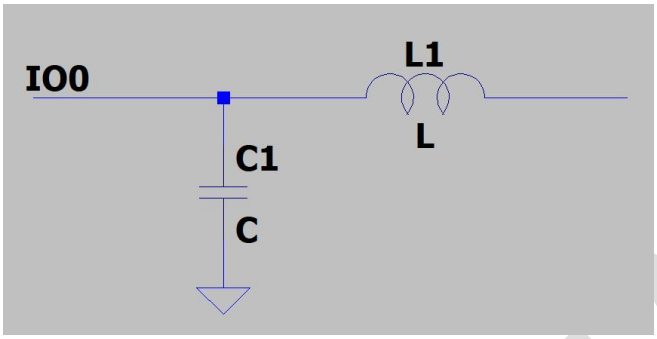
a. NA

## 1.7. I/O

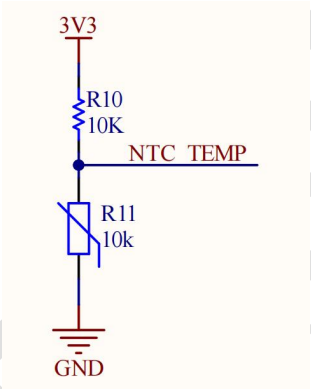
a. Please make sure input of GPIO will not be more than 3.6 V.

CONFIDENTIAL

b. If space is allowed, it is suggested to reserve a CL close to GPIO0 pin to improve EMC performance. Initially, do not stuff for C1 and use 0 ohm resistor for L1.

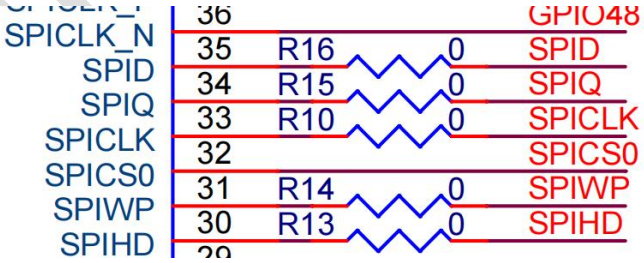



c. Please add a filtering cap close to ADC GPIO.




## 1.8. Flash/PSRAM

a. It is recommended to add a zero-ohm series resistor on the SPI communication lines for future adjustment.




## 1.9. UART

a. OK

## 1.10. Package

a. Which S3 chip are you using? S3R8 or S3FH4R2?

CONFIDENTIAL

## 1.11. Others

a. NA

CONFIDENTIAL

# 2. PCB

## 2.1. PCB Check

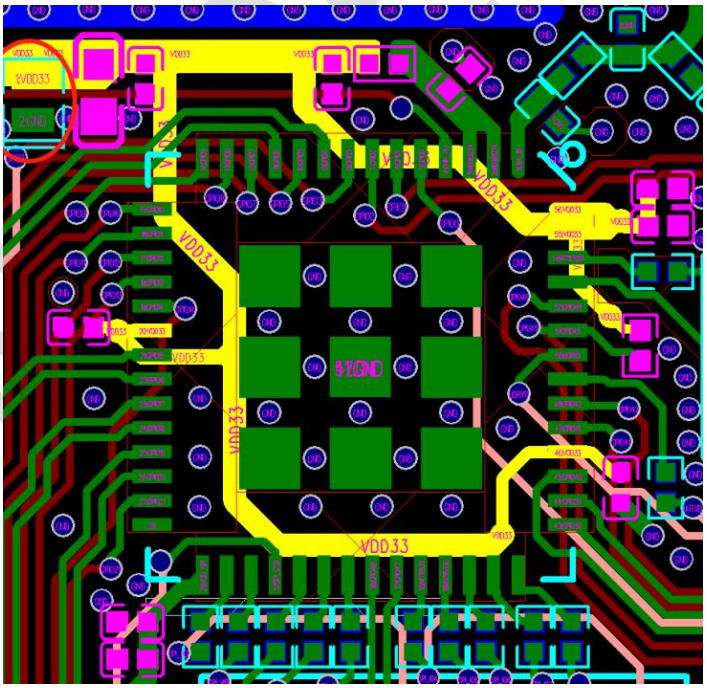
a. FR4 material is recommended for PCB layout.

b. It is recommended to use 4 or more layer design.

## 2.2. Power and Ground

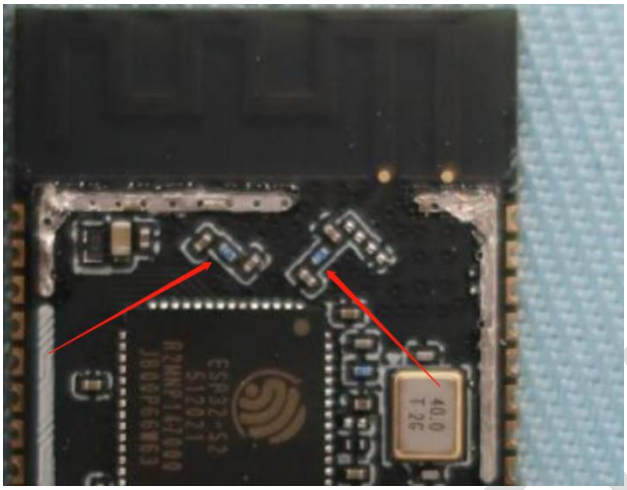
a. For qualified RF performance, please provide a complete ground for ESP32-S3 chip, RF and crystal part. It is why we do not recommend two-layer design since the ground will always be cut off. If you still want to use two-layer design, make sure you could solve this problem.

b. When using a four-layer board design, please follow the power supply layout of the module. First, the power traces should be routed in the inner layer (not the ground layer) . Then the trace width at the entrance should be at least 25 mil, the trace width for pins 3 and 4 should be at least 20 mil, and other power traces are recommended to be at least 10 mil.

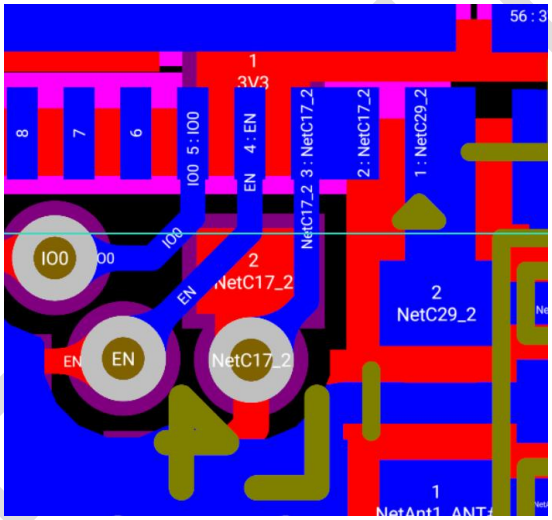



c. It is recommended to use 0201 package for capacitors of 1 uF and below, and then place the capacitors as close to the power pins as possible. Especially pins 2 and 3, the corresponding filter circuit needs to be as close to this pin as possible. You can refer to the figure below for this part. The power traces of pins 2 and 3 are routed at a 45° angle to the pins, and the RF traces next to them are routed at a 45° angle to the other side. In this way, the distance between the power supply and the RF can be as wide as possible, and the related circuits can be placed close to their respective pins.

CONFIDENTIAL

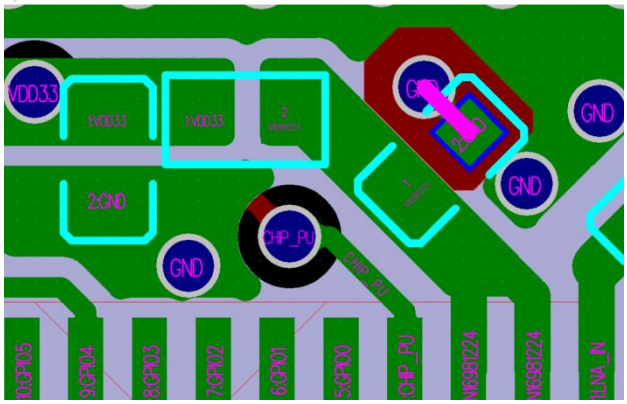


d. Please do not add via on pins 2 and 3 traces.



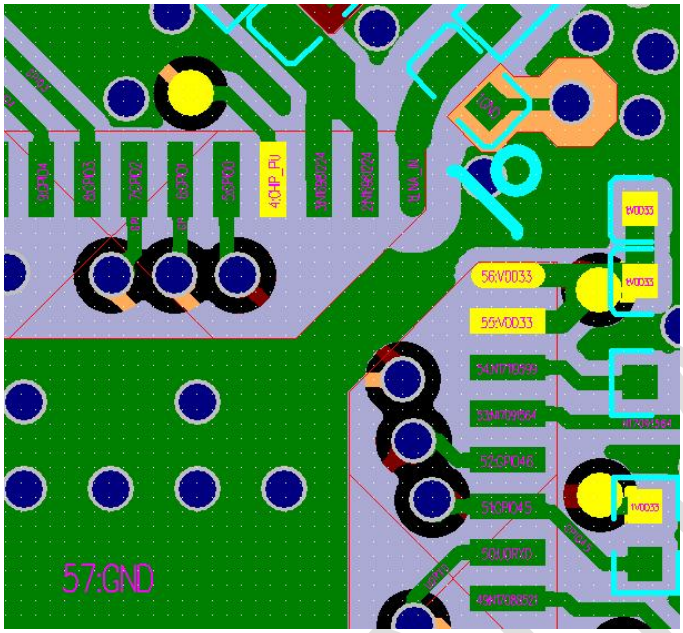

e. All power supplies, especially pins 2 and 3, must be adequately grounded and surrounded by dense ground holes, so that the harmonic suppression effect is better.

f. In the circuit of pins 2 and 3, you can add a via on the first capacitor through other layers to the bottom layer, which can further suppress harmonic interference.




g. It is suggested to connect S3 EPAD to external ground through traces or copper.

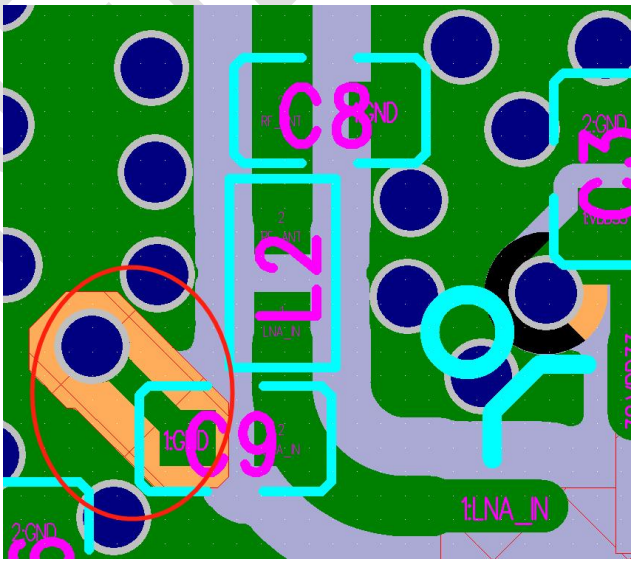
CONFIDENTIAL




## 2.3. RF

a. Please be noted that all RF traces should have 50 ohms impedance control in process.

b. Please use the 0201 package for CLC matching and place it as close to the chip pins as possible. The matching circuit is placed in a Z-shape, that is, the two capacitors should not face the same direction. Then please add a small stub at the first capacitor (close to the chip) to suppress harmonics. The length of the stub is 15 mil, the via is drilled to the third layer, and the impedance is approximately 100 Ω ± 10%.




c. There is too little ground for RF on the current layout, please improve this point, to make sure RF traces are surrounded by enough ground and dense via.

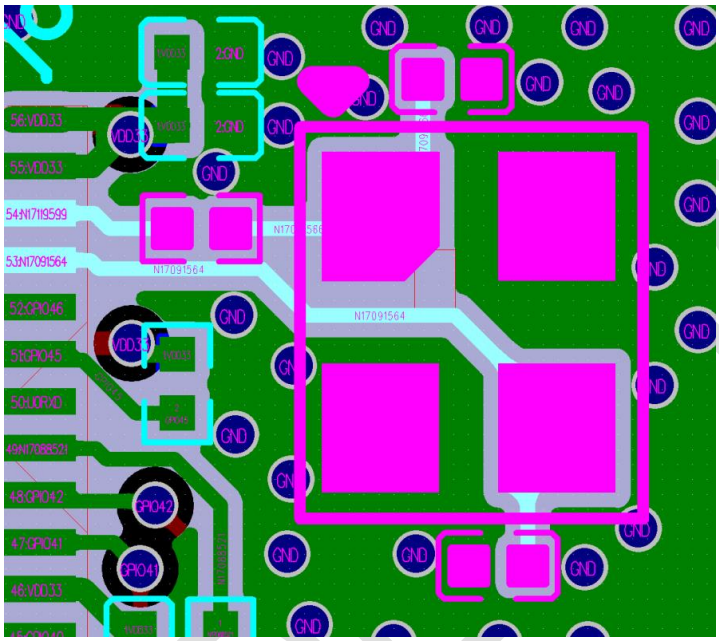
d. Please add another CLC to adjust antenna.

CONFIDENTIAL

e. Please make sure PCB antenna is simulated and tested.

## 2.4. Clock

a. There is too little ground for crystal part. Please improve crystal part referring to the figure below:




## 2.5. Reset/Enable

a. Enable traces are suggested to be surround by ground.

## 2.6. External RC

a. NA

## 2.7. I/O

a. NA

## 2.8. UART Layout

a. OK

## 2.9. Others

a. NA

CONFIDENTIAL

CONFIDENTIAL



Espressif IoT Team

*www.espressif.com*

Disclaimer and Copyright Notice

Information in this document, including URL references, is subject to change without notice.

THIS DOCUMENT IS PROVIDED AS IS WITH NO WARRANTIES WHATSOEVER, INCLUDING ANY WARRANTY OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR ANY PARTICULAR PURPOSE, OR ANY WARRANTY OTHERWISE ARISING OUT OF ANY PROPOSAL, SPECIFICATION OR SAMPLE.

All liability, including liability for infringement of any proprietary rights, relating to use of information in this document is disclaimed. No licenses express or implied, by estoppel or otherwise, to any intellectual property rights are granted herein.

The Wi-Fi Alliance Member logo is a trademark of the Wi-Fi Alliance. The Bluetooth logo is a registered trademark of Bluetooth SIG.

All trade names, trademarks and registered trademarks mentioned in this document are property of their respective owners, and are hereby acknowledged.

Copyright © 2023 Espressif Inc. All rights reserved.